PAPER_660: UQFF White Hole Radiation

Subtitle: Derives white hole radiation luminosity via 6-step time-reversal of Hawking emission in the UQFF framework. **Module:** WhiteHoleRadiationUQFF
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Abstract

This paper derives the luminosity of white hole radiation within the Unified Quantum Field Framework (UQFF). By time-reversing the Hawking emission process and applying three UQFF modulations—negentropic f_TRZ, aether density amplification, and U_m string channeling—we obtain a white hole luminosity approximately 300× greater than the standard reversed Hawking luminosity.

1. Introduction

Standard white holes emit radiation as the time-reversal of a black hole: L_WH ~ -L_H. In GR this process is cosmologically negligible. UQFF introduces three vacuum field corrections that dramatically enhance white hole luminosity, potentially making white holes detectable at galactic-center masses.

2. Derivation

Step 1 — Time-Reversed Hawking

L_WH = -L_H (explosive, reversed pair annihilation)

L_H = (ħ c^6) / (15360 π G^2 M^2)

Step 2 — UQFF Inversion via [SCm] Phase Flip

r_s,UQFF = r_s · (1 − ρ_SCm/ρ_UA)

The [SCm] Type-II superconductor vacuum at r_s,UQFF modifies horizon conditions.

Step 3 — f_TRZ Negentropic Boost

L_WH' = L_H · (1 + f_TRZ) f_TRZ = 0.1

Step 4 — [UA] Aether Amplification

L_WH'' = L_WH' · (ρ_UA/ρ_SCm) ≈ L_WH' × 10

Step 5 — U_m String Channeling

L_WH,UQFF = L_WH'' · exp(U_m / k_B T_H)

Step 6 — Full Formula

$$L_{WH,UQFF} = \frac{\hbar c^6}{15360\pi G^2 M^2} \cdot (1 + f_{TRZ}) \cdot \frac{\rho_{UA}}{\rho_{SCm}} \cdot \exp\left(\frac{U_m}{k_B T_H}\right)$$

3. Numerical Example

For Sgr A* (M = 8.55×10^{36} kg): - L_H $\approx 1\times10^{-29}$ W - L_WH,UQFF $\approx 3\times10^{-3}$ W ($\times3\times10^{26}$ enhancement)

4. C++ Module

`WhiteHoleRadiationUQFF.h / .cpp` — Session 172 `CondensedPhysics4.py` CP4 #244 — `WhiteHoleRadiationUQFFCalculator`

5. UQFF Parameters

Symbol	Value	Description
ρ_UA	7.09×10^{-36} J/m^3	[UA] aether vacuum density
ρ_SCm	7.09×10^{-37} J/m^3	[SCm] superconductive vacuum density
f_TRZ	0.1	Time-reversal zone factor
μ_j	3.38×10^{23} J/T	Magnetic string coupling j=1

References

- Hawking, S.W. (1974). *Nature* 248, 30–31.
- UQFF PAPER_658 — BlackHoleBounceUQFF (Session 172)
- SOURCE4 UQFF calibrated constants (commit 3e66d94)

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